

MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati

Semester: January-April, 2024

Tutorial Sheet No. 6

Date of Discussion: April 24, 2024 (Regular class hours)

Lectures 17-20: Systems of first-order equations, stability + extra discussion on Bessel functions and Legendre polynomials

1. The fox-rabbit problem is given by the following system:

$$\begin{cases} \frac{dx}{dt} = x(A - By), \\ \frac{dy}{dt} = -y(C - Dx), \end{cases}$$

where each of A, B, C, D is given and they have their own meaning. By eliminating y from the above system, obtain the nonlinear second-order equation in x .

2. Find the critical points of the following nonlinear systems:

(a) $\frac{d^2x}{dt^2} + \frac{dx}{dt} - (x^3 + x^2 - 2x) = 0;$

(b) $\begin{cases} \frac{dx}{dt} = y^2 - 5x + 6, \\ \frac{dy}{dt} = x - y. \end{cases}$

3. Describe the phase portrait of the following systems:

(a) $\begin{cases} \frac{dx}{dt} = x, \\ \frac{dy}{dt} = 0. \end{cases}$

(b) $\begin{cases} \frac{dx}{dt} = -x, \\ \frac{dy}{dt} = -y. \end{cases}$

4. Determine the nature and stability of the critical point $(0, 0)$ for the following autonomous systems:

(a) $\begin{cases} \frac{dx}{dt} = 2x, \\ \frac{dy}{dt} = 3y. \end{cases}$

(b) $\begin{cases} \frac{dx}{dt} = -3x + 4y, \\ \frac{dy}{dt} = -2x + 3y. \end{cases}$

(c) $\begin{cases} \frac{dx}{dt} = -4x - y, \\ \frac{dy}{dt} = x - 2y. \end{cases}$

(d) $\begin{cases} \frac{dx}{dt} = 4x - 2y, \\ \frac{dy}{dt} = 5x + 2y. \end{cases}$

5. The free vibrations of a mass attached to a spring is governed by the following second-order differential equation:

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + a^2x = 0,$$

- where $b \geq 0$ and $a > 0$ are appropriate constants related to the medium and the spring.
- Convert it to an equivalent system of first-order equations.
 - Find the auxiliary equation of the system.
 - Describe the nature and stability of the critical point for the cases (i) $b = 0$, (ii) $0 < b < a$, (c) $b = a$, (d) $b > a$.

(Extra problems on Bessel functions and Legendre polynomials)

6. Verify that $xJ_0(x)$ is a solution of $y'' + y = -J_1(x)$, $x > 0$. Using it, show that

$$\int_0^x \cos(x - \xi) J_0(\xi) d\xi = xJ_0(x), \quad x > 0.$$

7. Making the substitution $y = xv(x)$, find the solution of the following differential equation in terms of Bessel functions:

$$xy'' - y' + xy = 0.$$

8. Making the substitution $y = v(x)/x$, find the solution of the following differential equation in terms of Bessel functions:

$$x^2y'' + 3xy' + (m^2x^2 + 1)y = 0, \quad m > 0.$$

9. If $P_n(x)$ is Legendre polynomial of degree n , and if a is such that $P_n(a) = 0$, show that $P_{n-1}(a)$ and $P_{n+1}(a)$ are of opposite signs.

10. Show that $\int_{-1}^1 xP_n(x)P_{n-1}(x) dx = \frac{2n}{4n^2 - 1}$.

11. Using Legendre polynomials, evaluate $\int_{-1}^1 \frac{x^2 dx}{\sqrt{5 - 4x}}$.

Note: It is not to be expected that all the problems will be discussed.

The End
